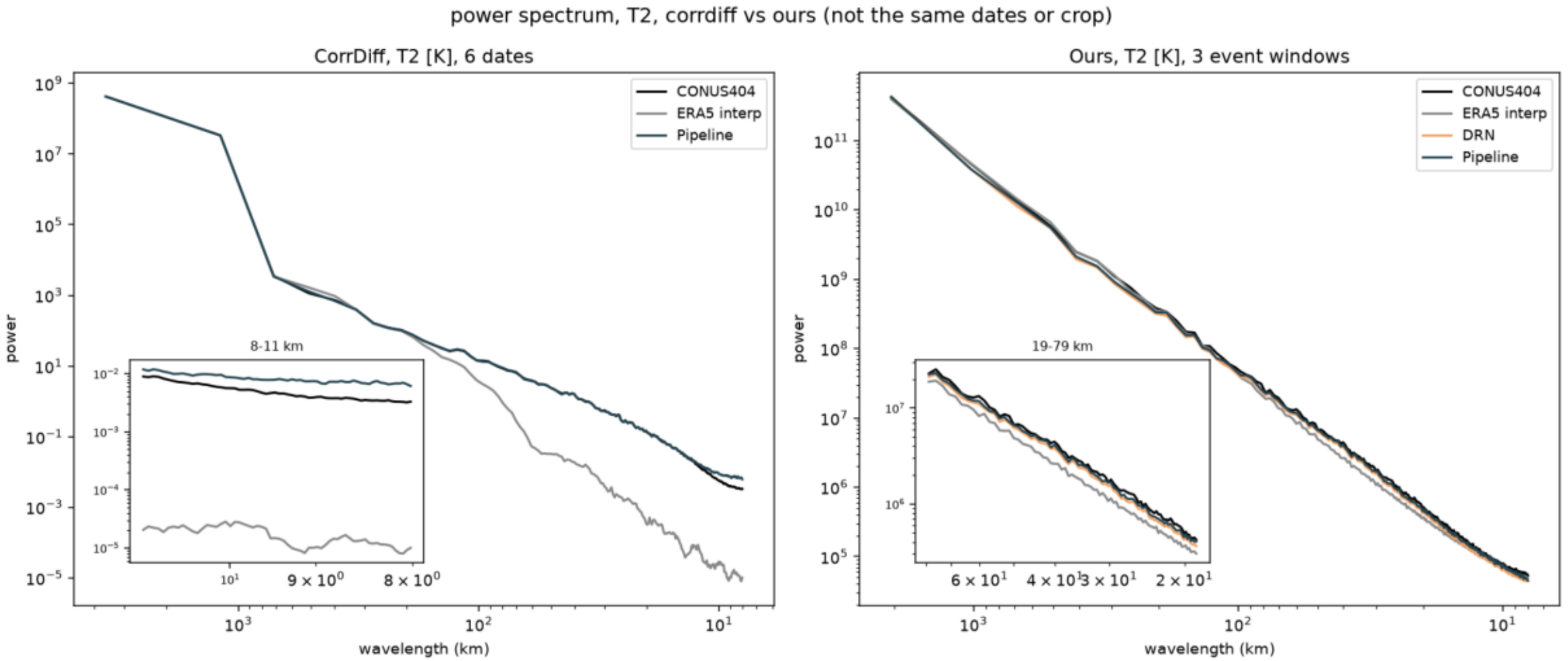
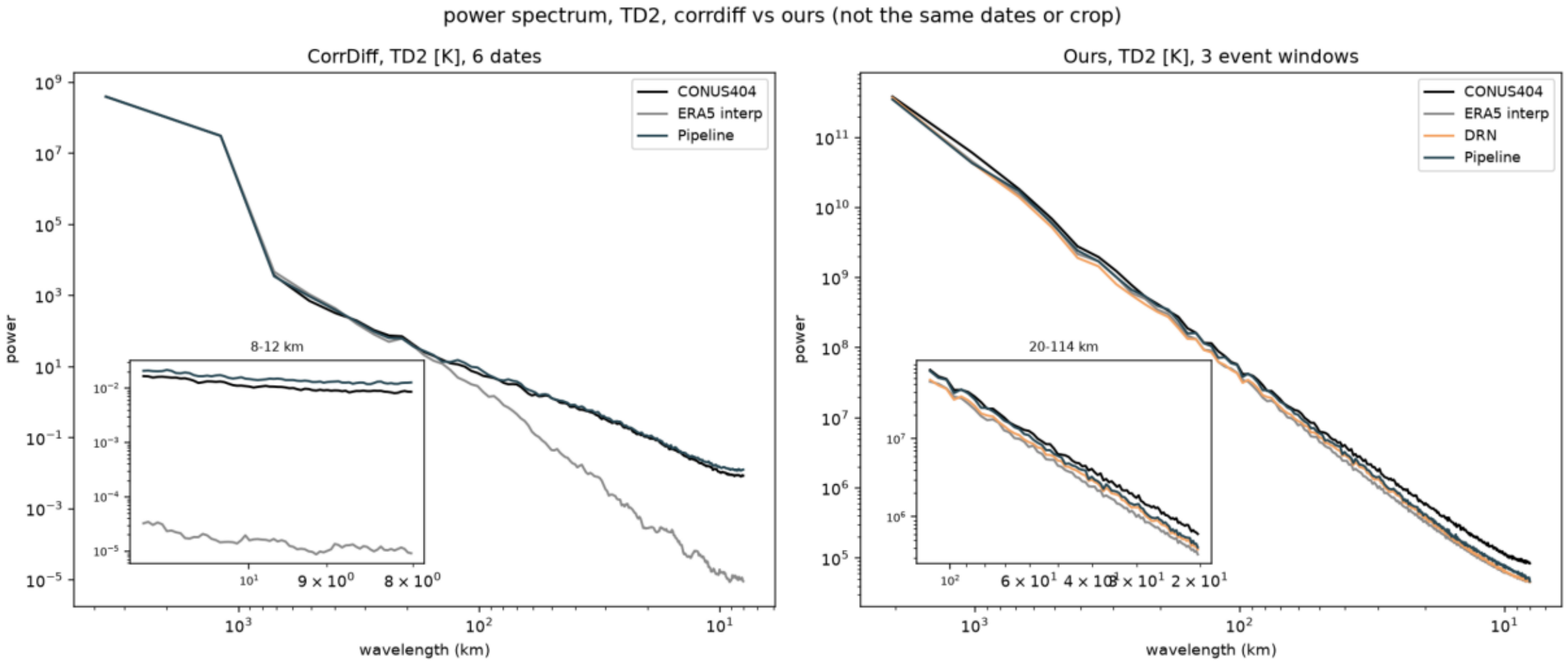
ERA5→CONUS404 Latent CorrDiff

Frequency-domain detail, CorrDiff comparison, Anderson Creek Fire case study

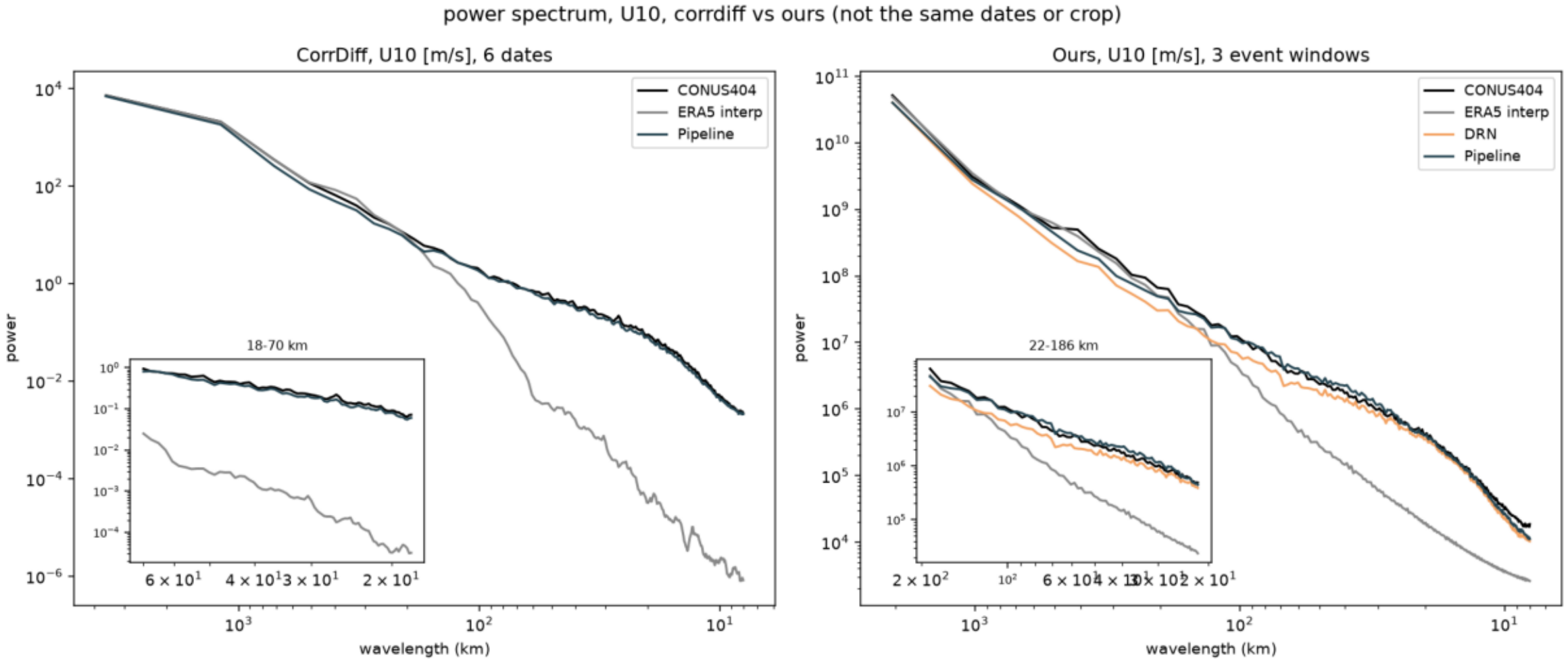
power spectrum, T2, corrdiff vs ours



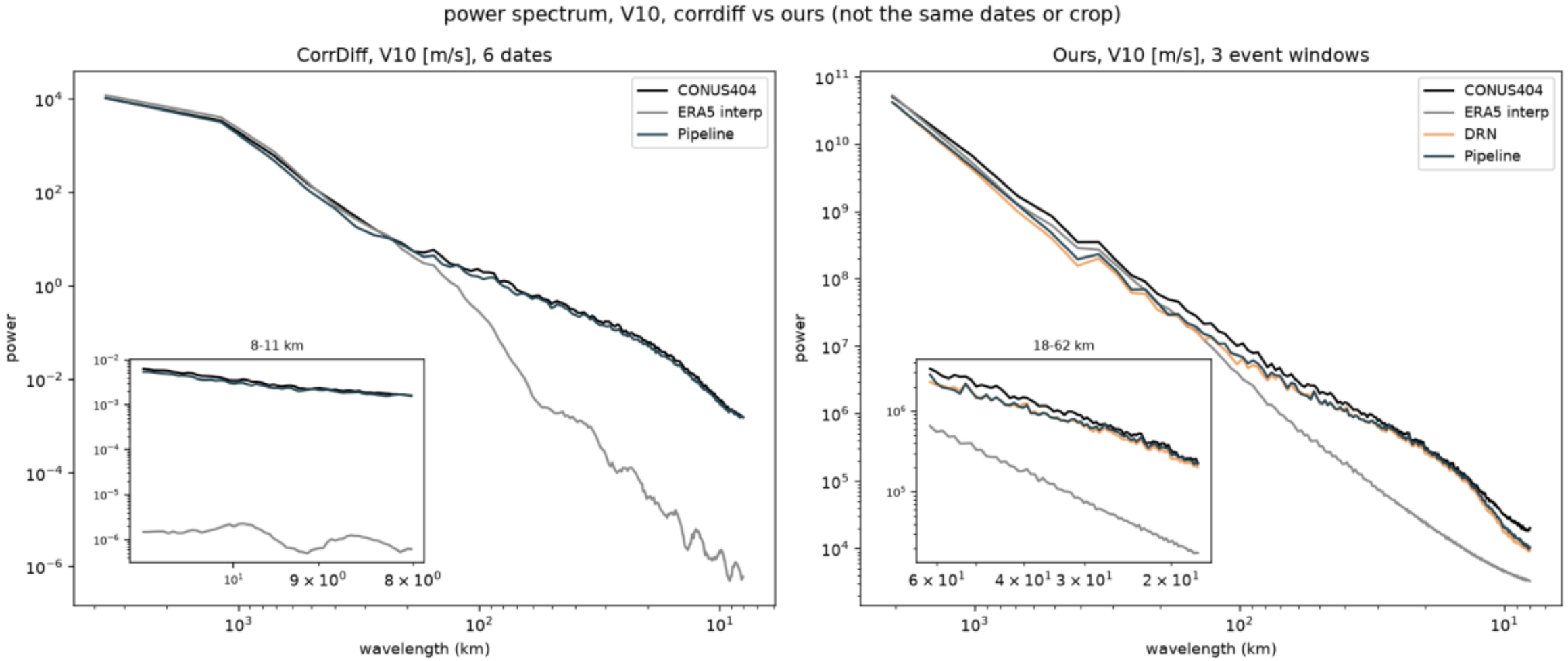
power spectrum, TD2, corrdiff vs ours



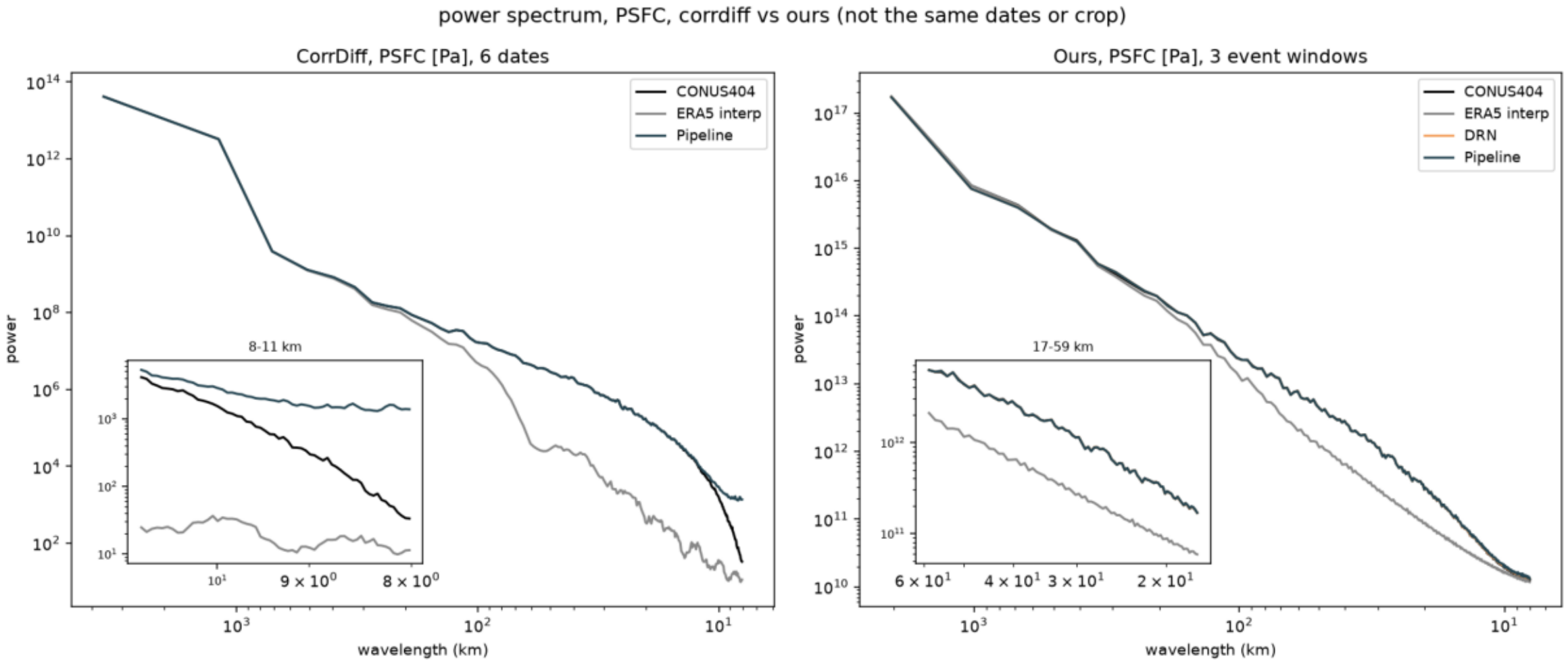
power spectrum, U10, corrdiff vs ours



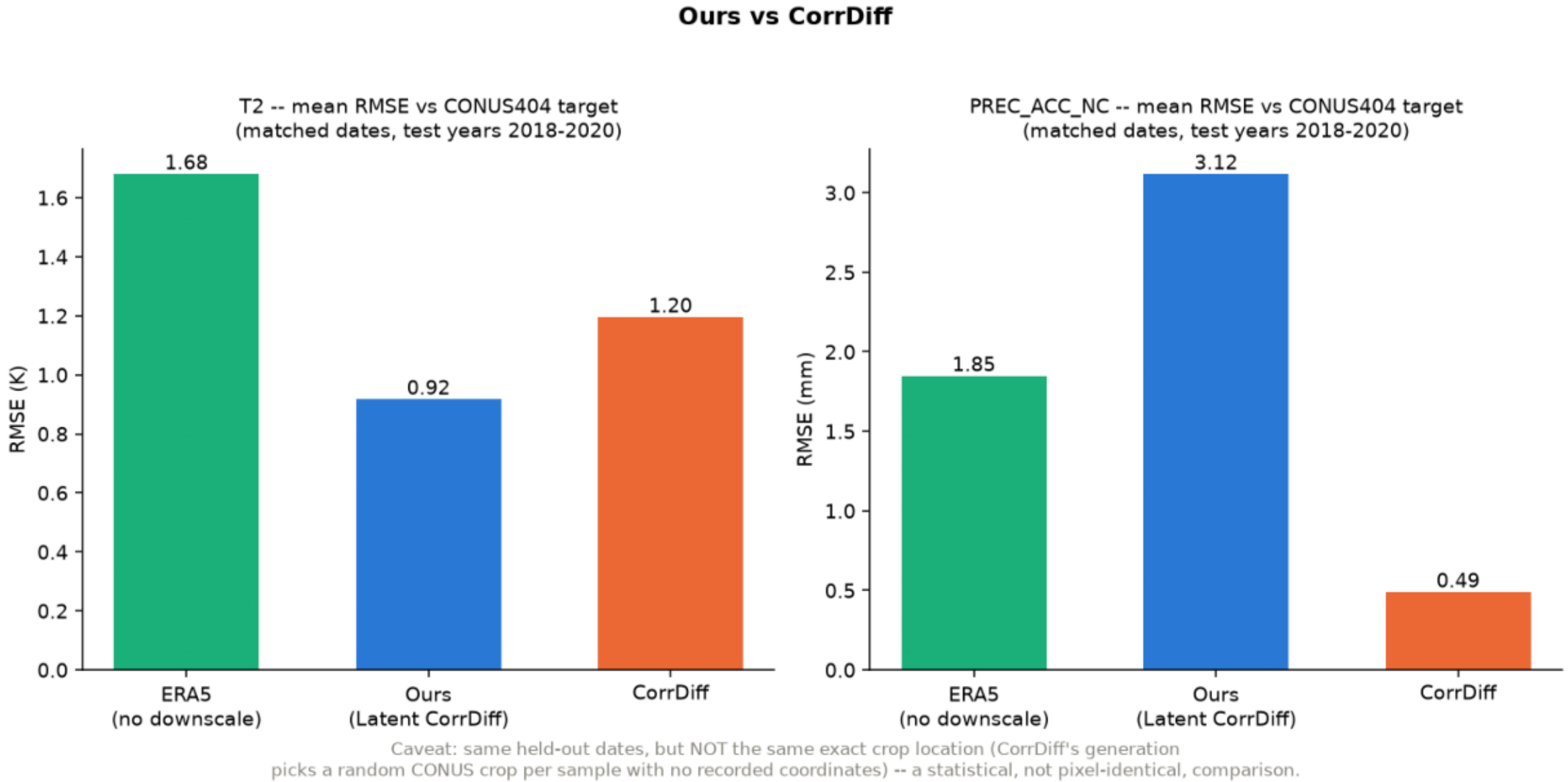
power spectrum, V10, corrdiff vs ours



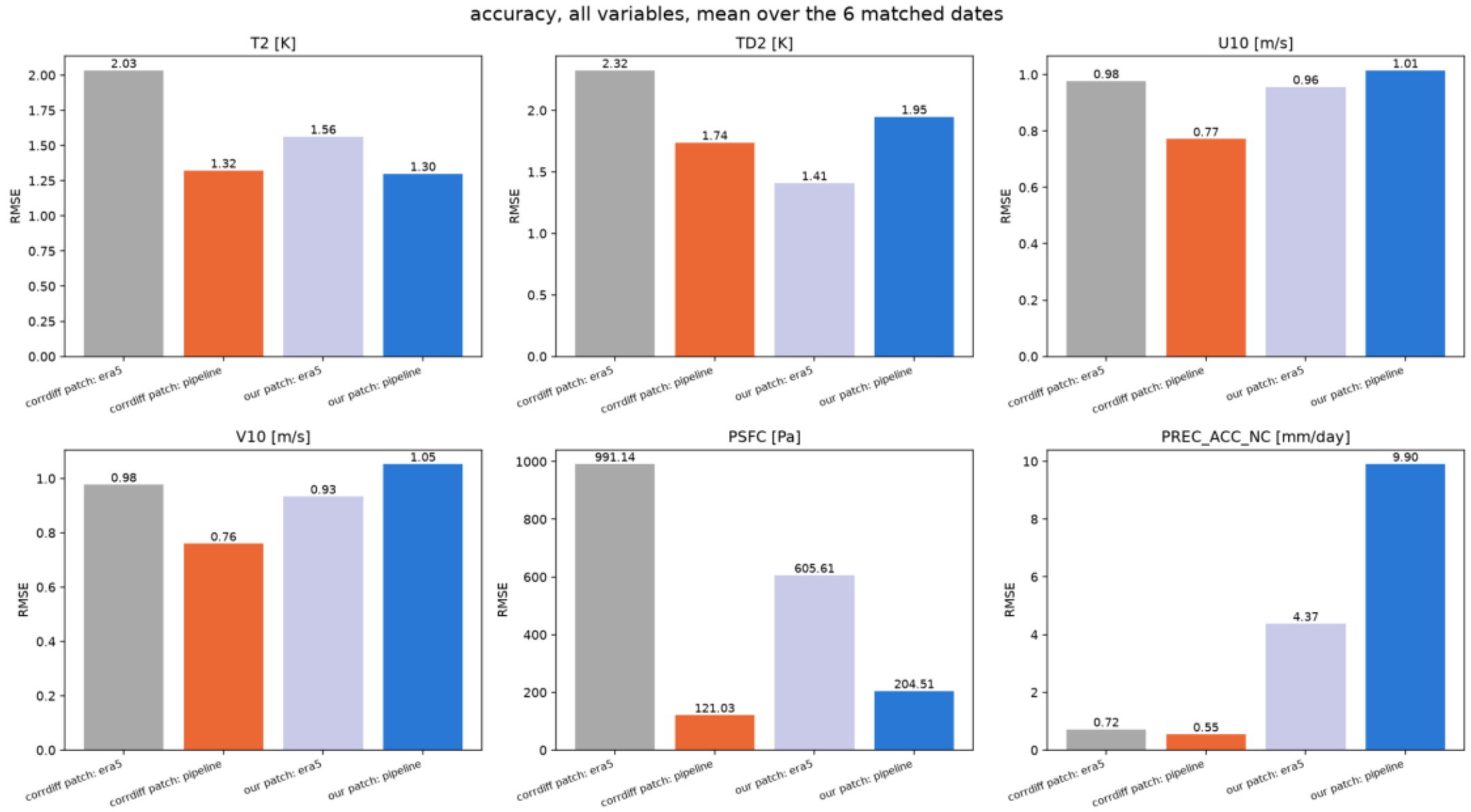
power spectrum, PSFC, corrdiff vs ours



ours vs corrdiff, T2 and precip only, mean RMSE



accuracy, all variables, corrdiff vs ours



CorrDiff training time, one clean pass

Regression (CorrDiffRegressionUNet, 5,000,000 steps): 7 SLURM jobs, 3 days 7 hours 39 minutes compute, about 318.6 GPU-hours on 4xA100

Diffusion (EDMPrecondSuperResolution, 5,000,000 steps): 10 SLURM jobs, 4 days 23 hours 2 minutes compute, about 476.1 GPU-hours

Total: 17 jobs, 8 days 6 hours 40 minutes compute, about 794.7 GPU-hours

Wall clock span: 8 days 18 hours 11 minutes (longer than compute time because of two multi hour SLURM queue waits between resubmissions)

Our training time (DRN + VAE + diffusion), clean path only

DRN: 17 jobs, 3 days 9 hours 29 minutes compute, about 326 GPU-hours

VAE: 4 jobs, 1 day 13 hours 17 minutes compute, about 149 GPU-hours

Diffusion: 16 jobs, 6 days 1 hour 37 minutes compute, about 582 GPU-hours (only reached epoch 106 of a planned 300)

Total: 37 jobs, 11 days 0 hours 23 minutes compute, about 1,058 GPU-hours

This is more total GPU-hours than CorrDiff's 794.7, even on the clean path, because our pipeline has three stages (DRN, then VAE, then diffusion) against CorrDiff's two (regression, then diffusion)

Wall clock: training time vs inference time point in different direct

Training (previous two pages): ours uses more total GPU hours (about 1,058 vs CorrDiff's 794.7), because our pipeline has three trained stages against CorrDiff's two

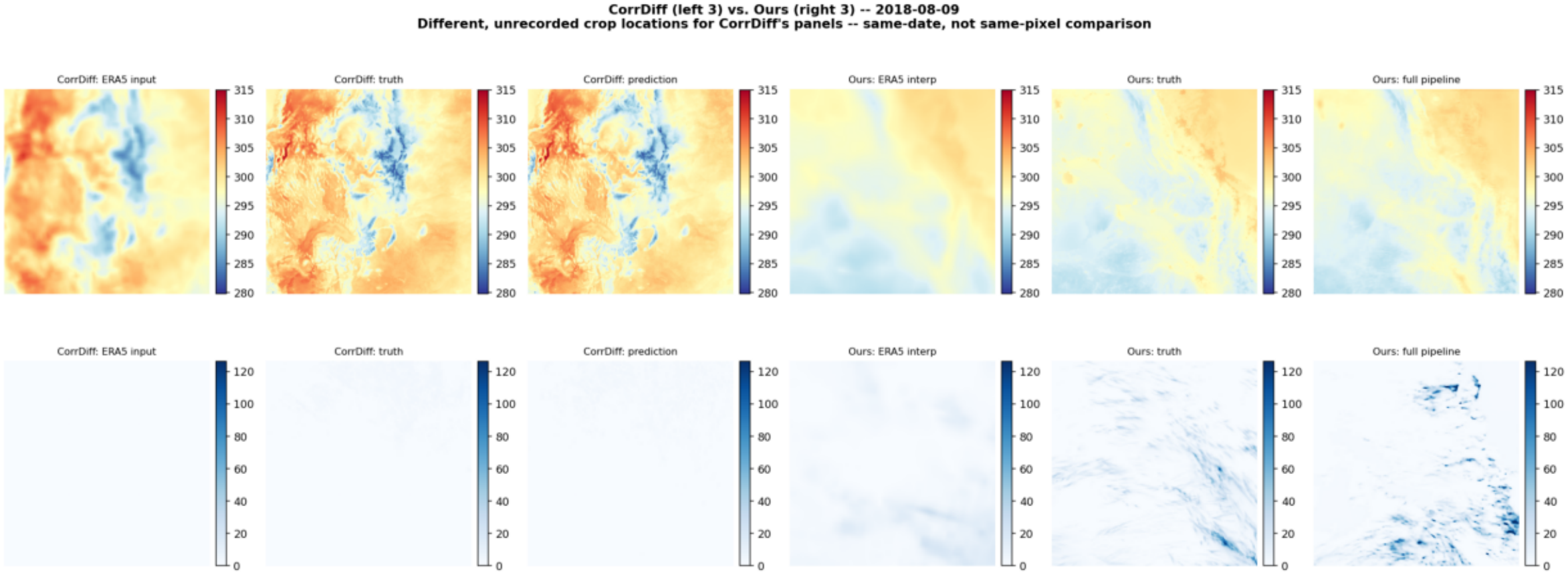
Inference, live example just run: ours took 1.03s for 4 diffusion steps and 4.25s for 16 steps (4 ensemble members, one A100, 256x256 patch)

CorrDiff's own logged inference timing (job 58824533): 64.6s total for 4 steps and 4 members on a 448x448 crop, about 4.04s per batch element

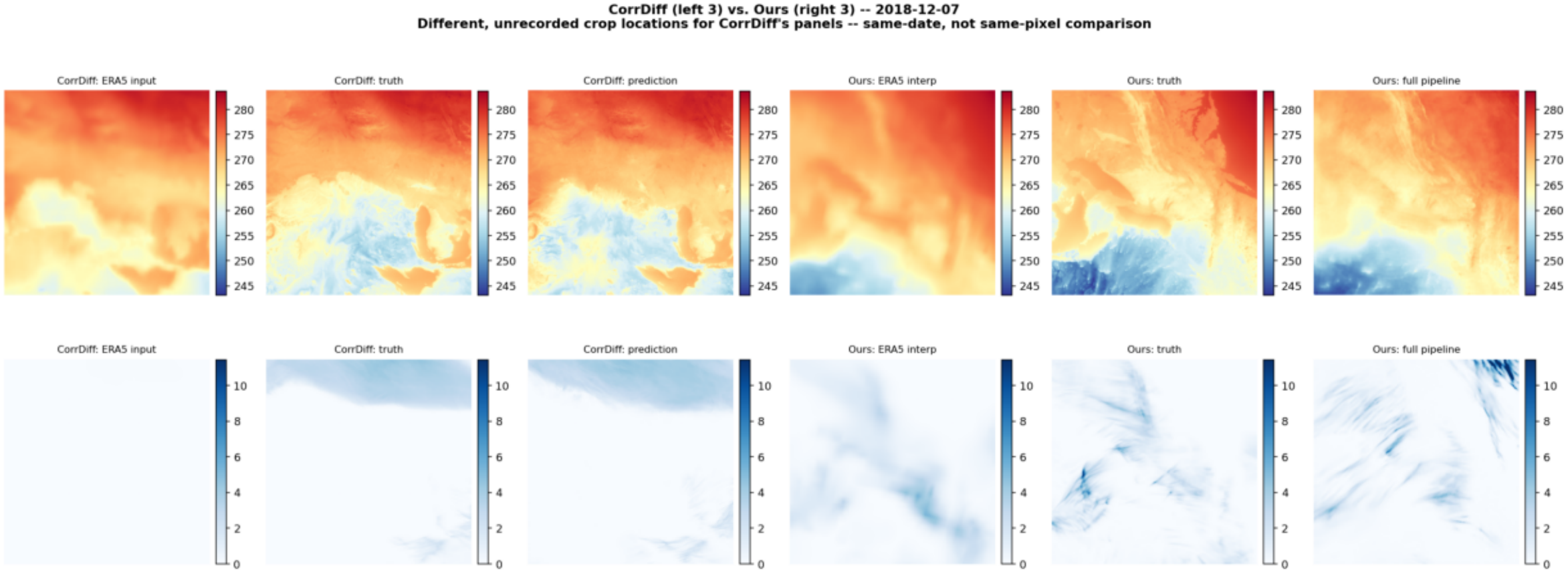
At matched step count (4 steps), ours was about 1s vs CorrDiff's about 4s per element, but the patches are not the same size, so part of that gap is resolution, not purely the architecture

Training cost and inference cost are two separate questions here and they do not agree: more trained stages cost more to train, but the smaller compressed latent space costs less to run per inference call

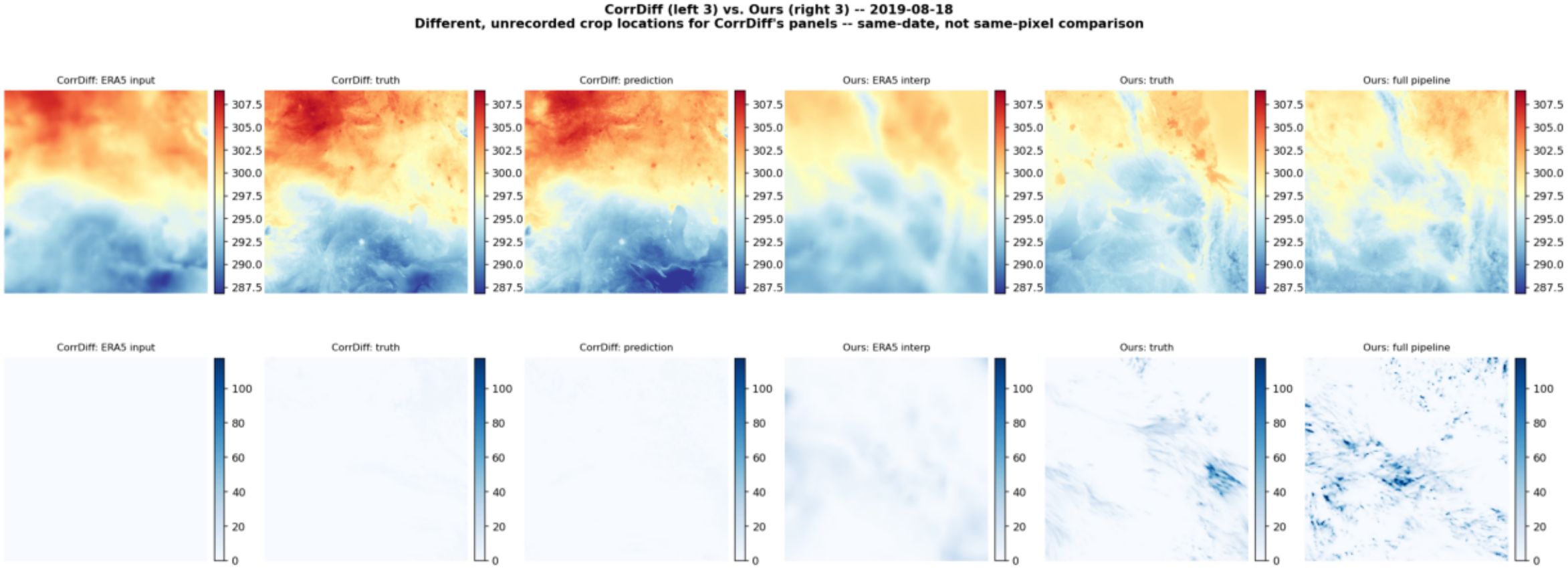
corrdiff vs ours, 2018-08-09, same date, different map locations



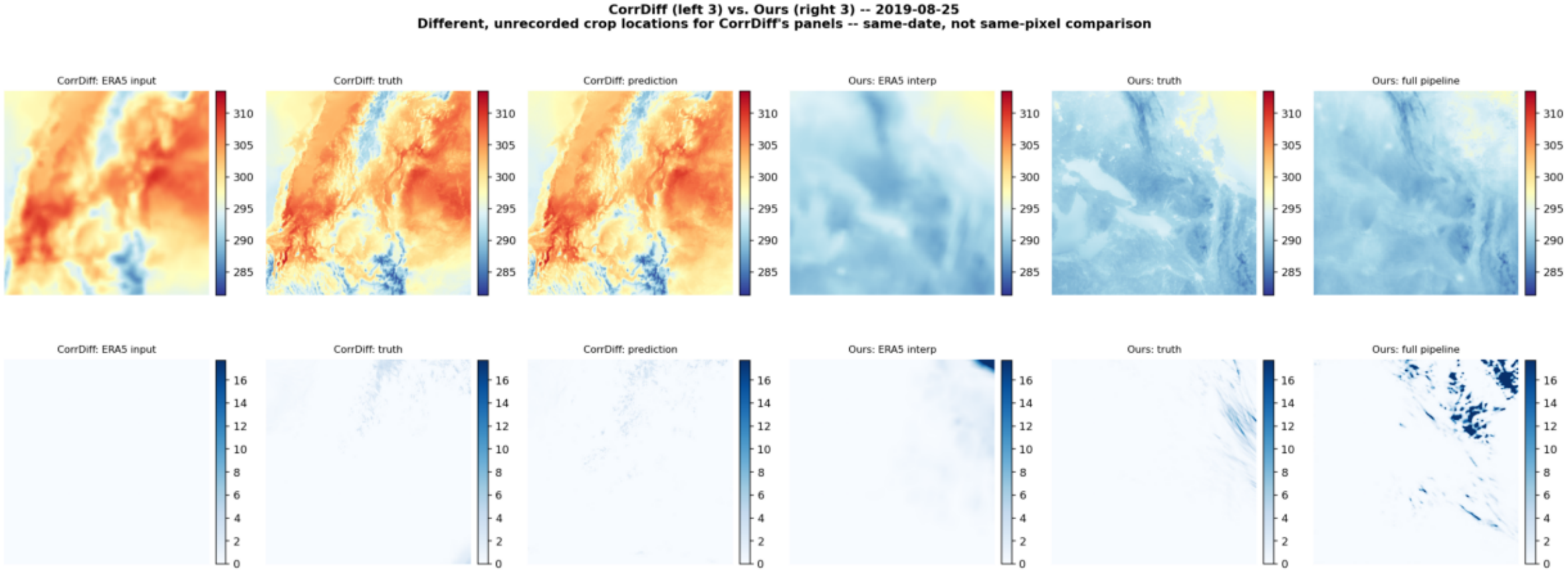
corrdiff vs ours, 2018-12-07, same date, different map locations



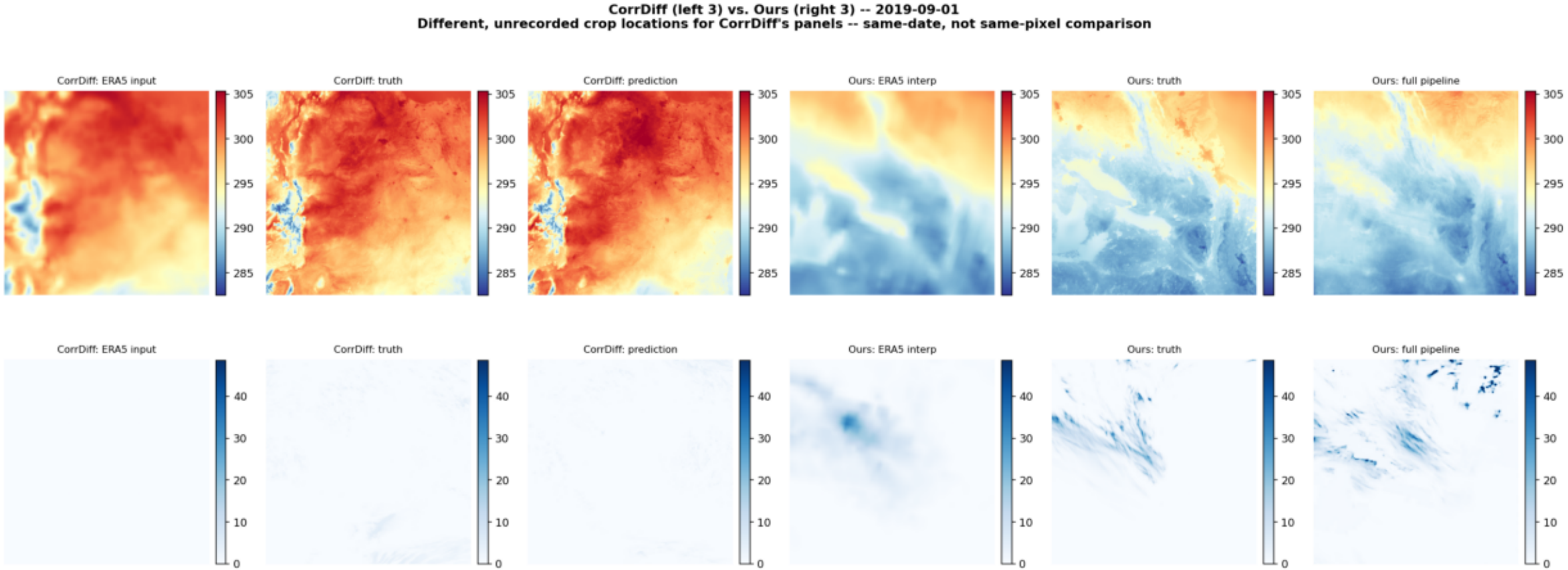
corrdiff vs ours, 2019-08-18, same date, different map locations



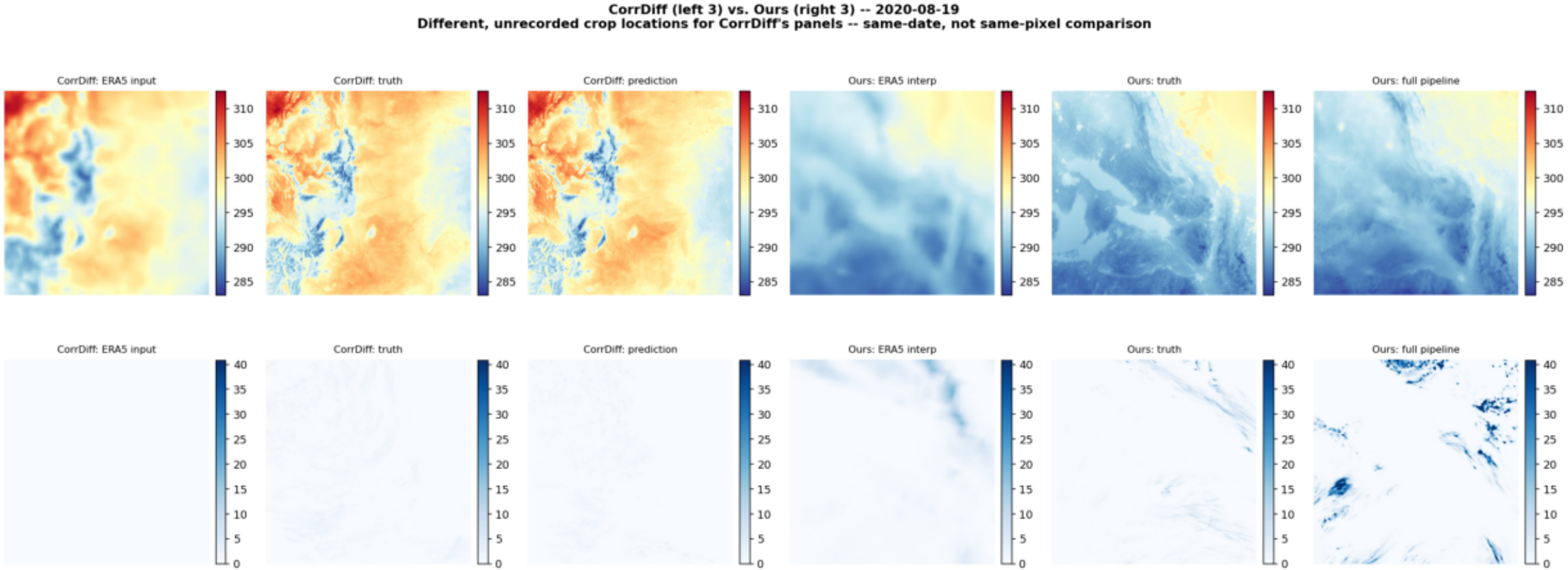
corrdiff vs ours, 2019-08-25, same date, different map locations



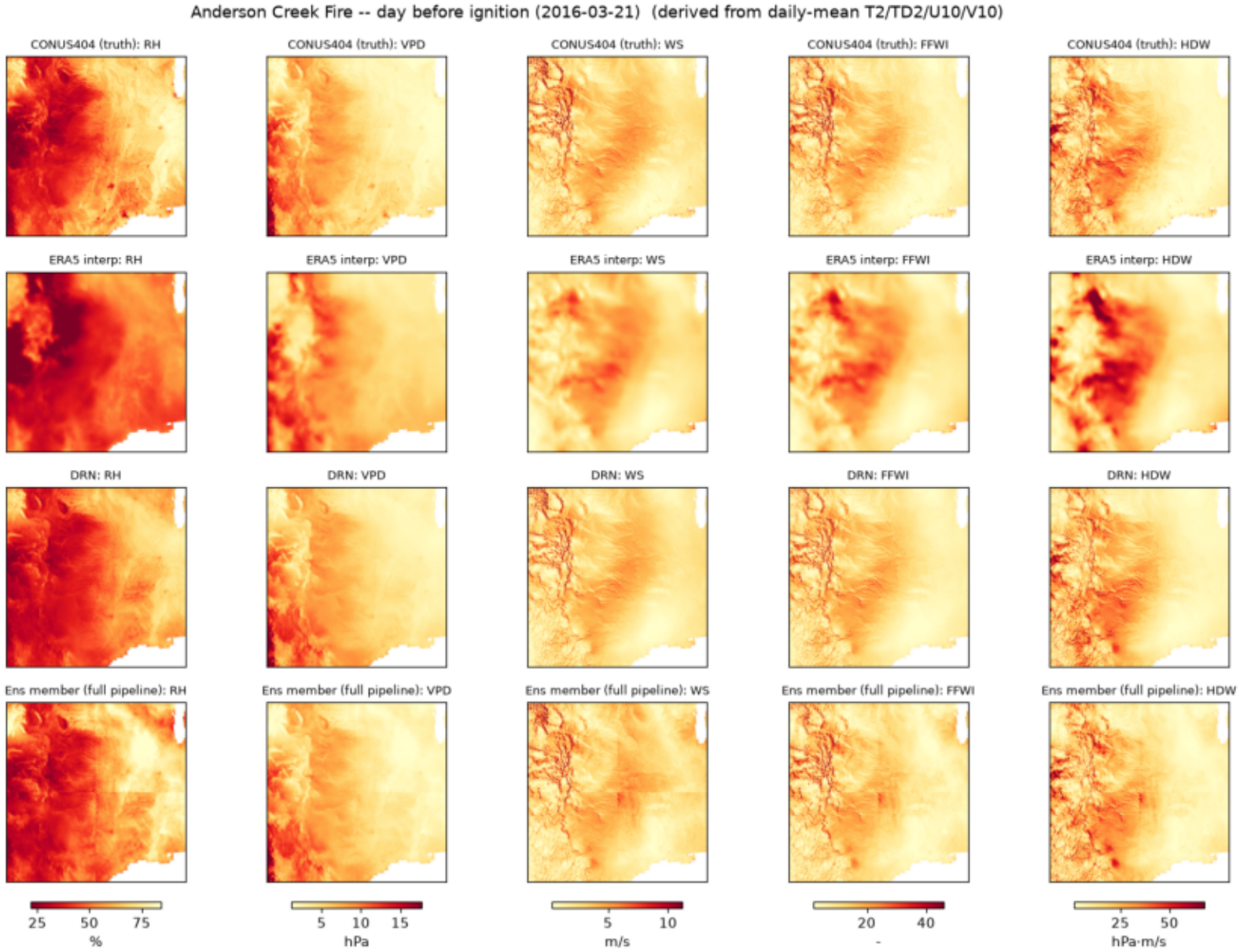
corrdiff vs ours, 2019-09-01, same date, different map locations



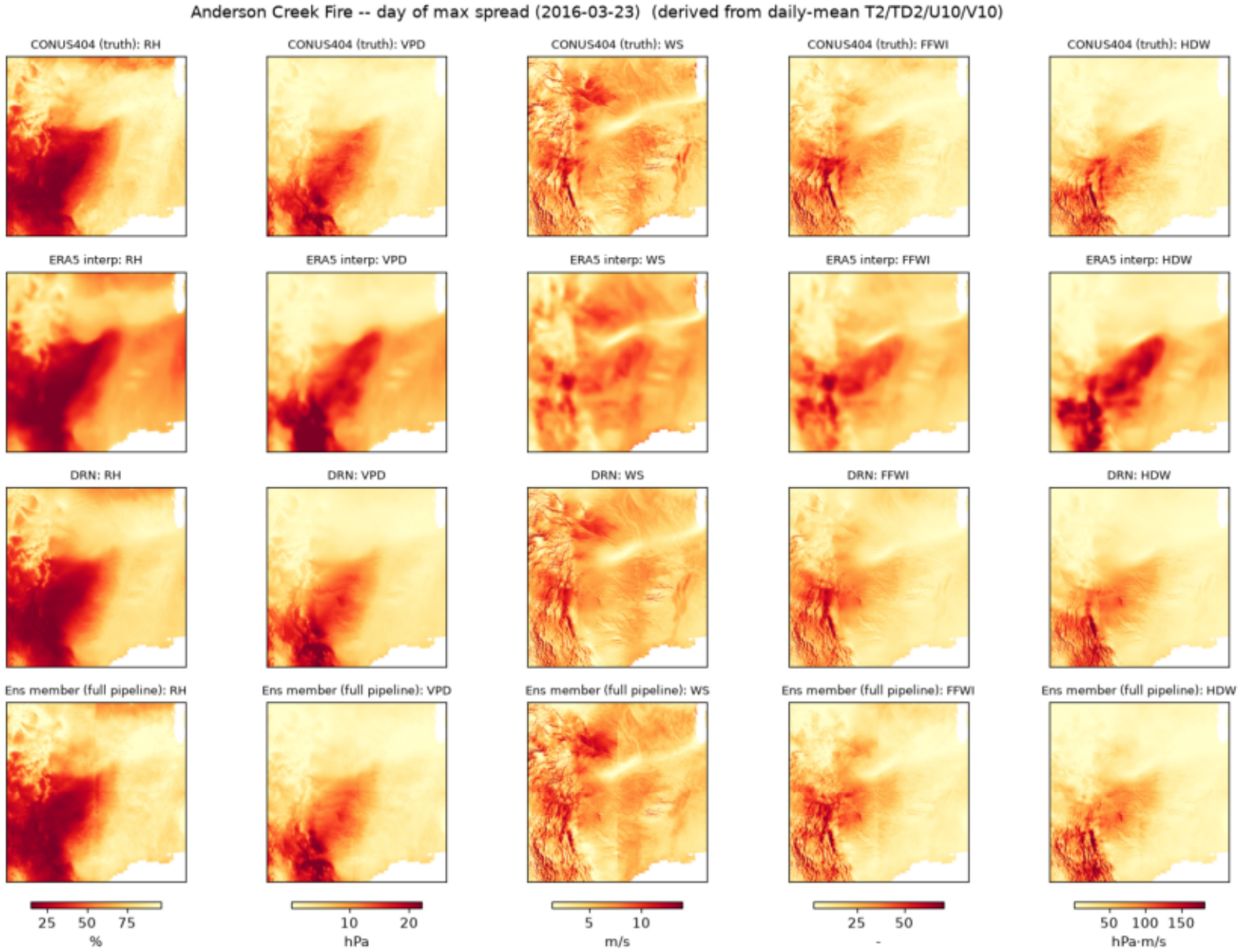
corrdiff vs ours, 2020-08-19, same date, different map locations



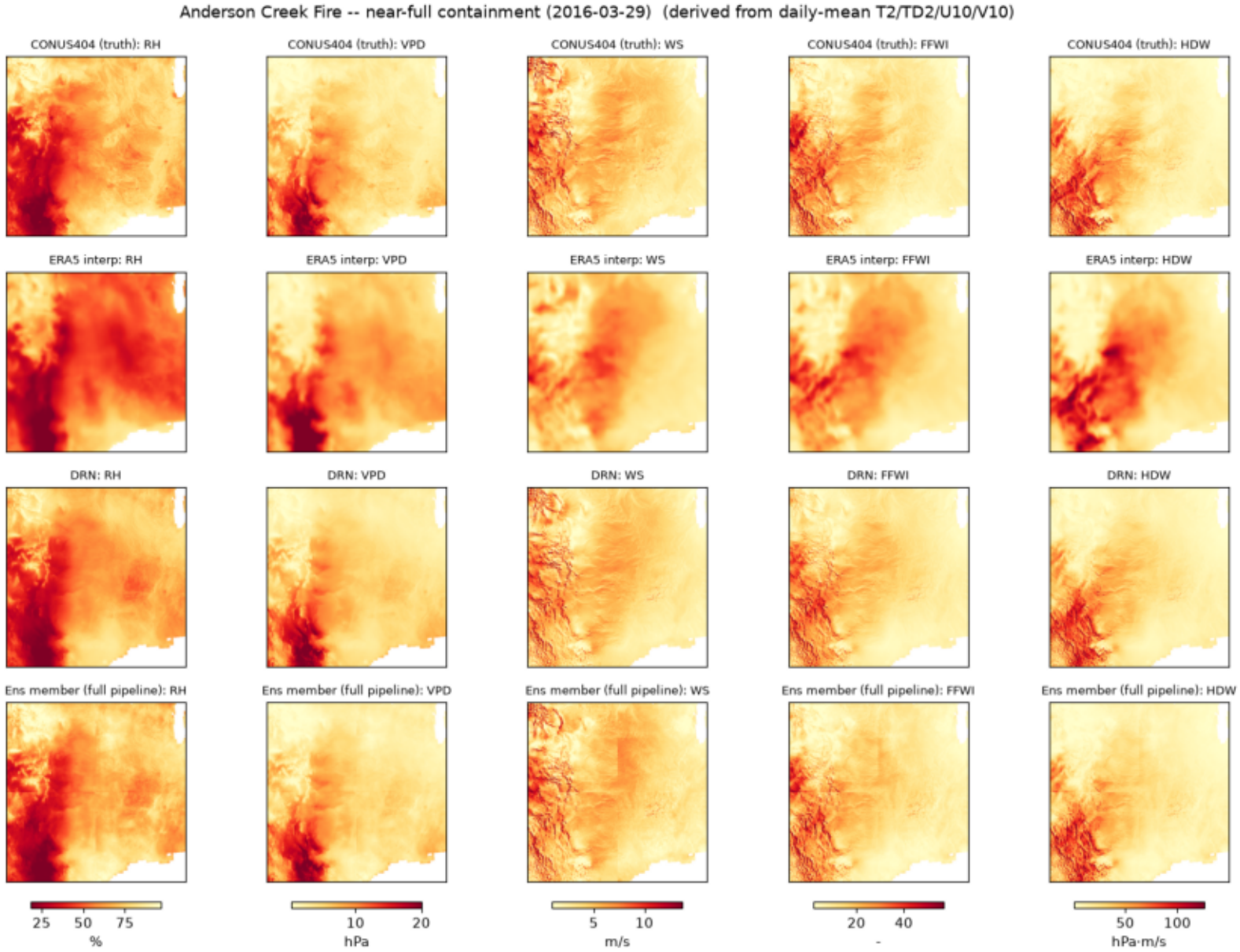
Anderson Creek Fire, day before ignition (2016-03-21)



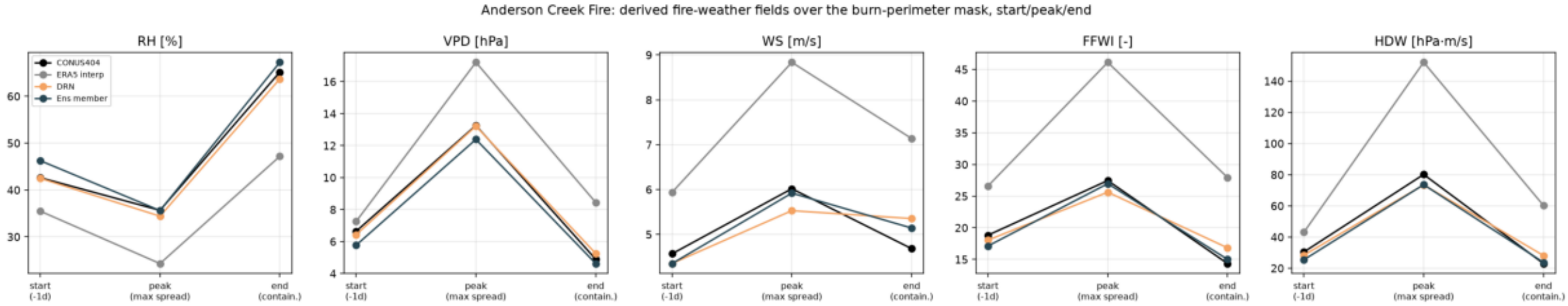
Anderson Creek Fire, day of maximum spread (2016-03-23)



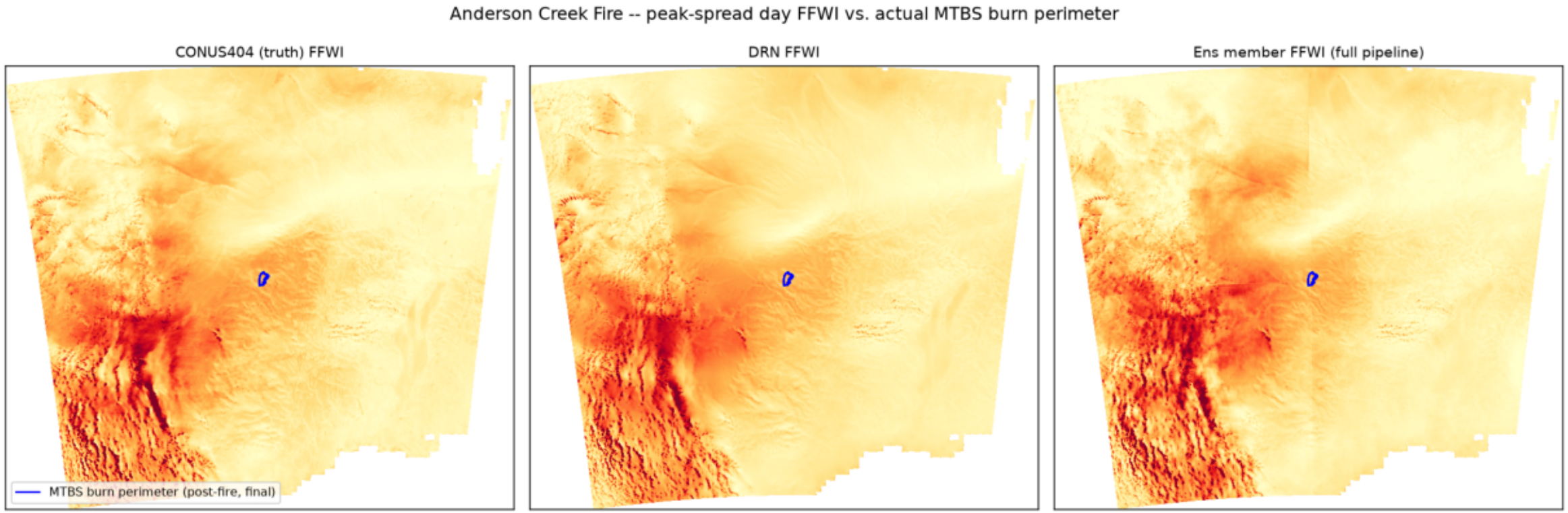
Anderson Creek Fire, near-full containment (2016-03-29)



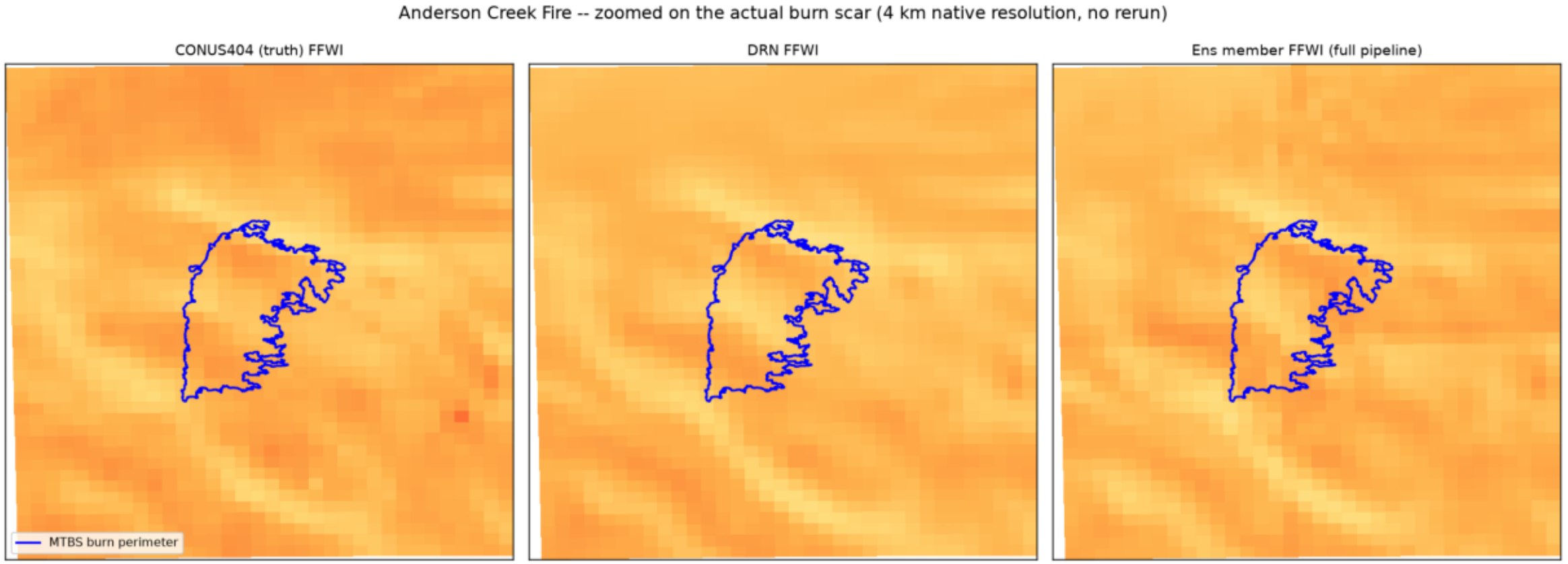
Fire-weather timeline: start → peak → containment



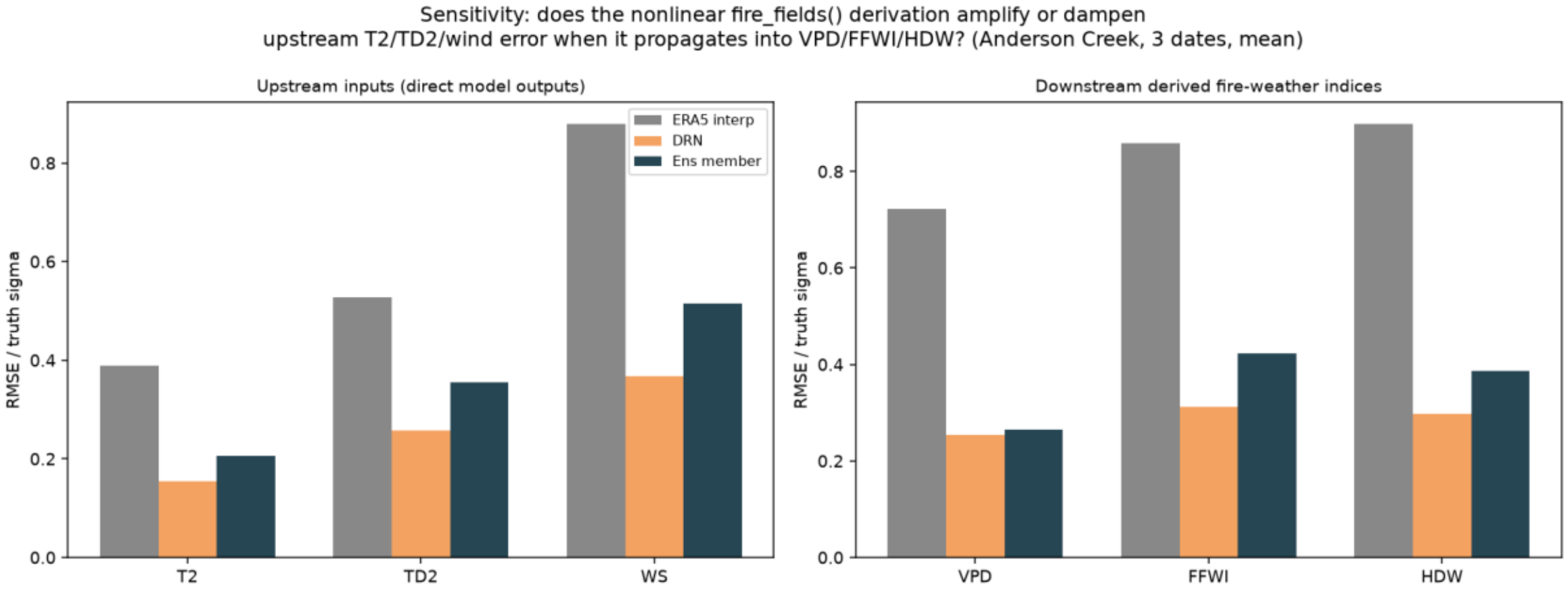
Peak-spread FFWI vs. the actual MTBS burn perimeter



Zoomed on the actual burn scar (4 km native resolution)



Sensitivity: does the fire-weather formula amplify upstream error?



Independent station check (AVK ASOS, ~46 km from the fire) vs. model

